# Nash equilibrium

Rock Paper Scissors Lizard Spock, all the way down

**Play**
Round robin, until one champion is left.

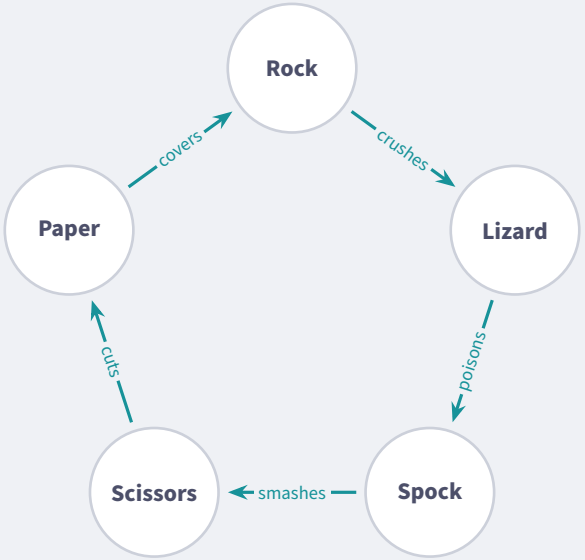
**Shrink**
Three actions instead of five.

**Solve**
Find the mix nobody can beat.

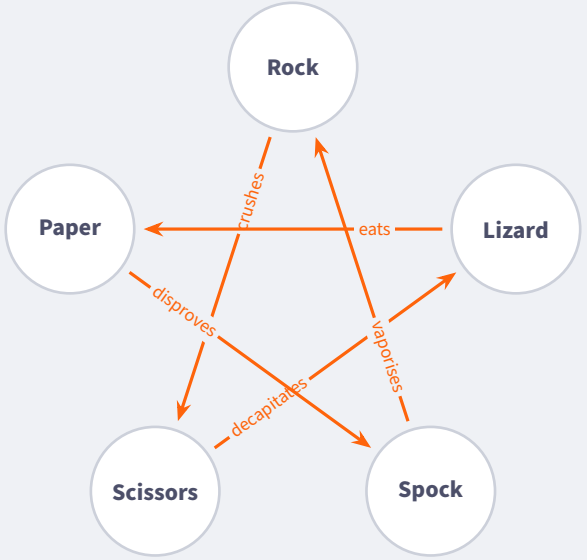
Class champion:

Did I beat them?

## Beat the one beside you




## Beat the one across from you




Each action wins ____, loses ____ and draws ____. So the game is _____, and every cell's payoffs add to ____.

| | Rock | Paper | Scissors |
|---|---|---|---|
| **Rock** | | | |
| **Paper** | | | |
| **Scissors** | | | |

Fill in the *row* player's payoff:

win = +1, draw = 0, lose = −1.

Then circle each player's best responses.

The column player's matrix is _____.

A cell with both players' best responses circled would be a pure Nash equilibrium.

Whatever the pair of actions, somebody wants to move:

If they play Rock, I want

If they play Paper, I want

If they play Scissors, I want

… and then they want to move too.

**So any equilibrium has to be _____.**

A mixed strategy $\sigma$ is a best response to $\tau$ if and only if



In words: every action I actually use has to be _____.

So if I mix over all three actions, all three must give me the same expected payoff.

Suppose the column player uses $\sigma_c = (x, y, 1 - x - y)$ over (Rock, Paper, Scissors).

$u_r(\text{Rock}, \sigma_c) =$

$u_r(\text{Paper}, \sigma_c) =$

$u_r(\text{Scissors}, \sigma_c) =$

Set them equal, and solve:

$\sigma_c =$

Support enumeration checks every pair of supports in turn.

Non-empty supports for one player, out of 3 actions:

Pairs of supports to check:

And with 5 actions, as in the game we played:

Which is why we want theorems, not enumeration.

What we just did in the room is **Question 1** on the Nash Equilibrium page.

| | | |
|---|---|---|
| **(a)** | Why it is symmetric and zero sum, and the column player's matrix. | [3] |
| **(b)** | Utilities for three given pairs of mixed strategies. | [3] |
| **(c)** | Use the best response condition to prove $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an equilibrium. | [5] |
| **(d)** | Prove it is the *unique* equilibrium. | [5] |
| **(e)** | State and justify the equilibrium for the five-action game. | [3] |

vknight.org/gt/topics/nash-equilibrium.html

Every stage of today, and the part of the notes it was reading.

| What we did | Notes | In the textbook |
|---|---|---|
| Five actions, then three | 2.2.3 | Games: Definition: Normal Form Game |
| Nobody wants to stand still | 5.2.1 | Nash Equilibrium: Definition: Nash Equilibria |
| Equal payoffs across the support | 3.2.6 | Rationalisation: Theorem: Best Response Condition |
| Checking every pair of supports | 5.2.3 | Nash Equilibrium: Definition: Support Enumeration Algorithm |
| **Examinable** | 3.2.6 | Theorem: Best Response Condition, and proof |

vknight.org/gtb/chapters/rationalisation, vknight.org/gtb/chapters/nash-equilibrium

Quick quiz before you go: vknight.org/gt/topics/nash-equilibrium.html

Before the next class: read the Nash Equilibrium chapter as far as support enumeration, and attempt Question 1.